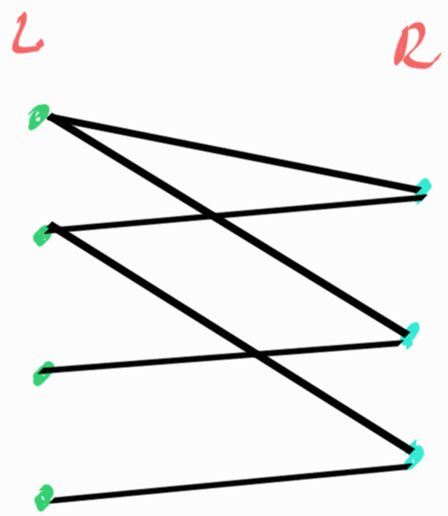
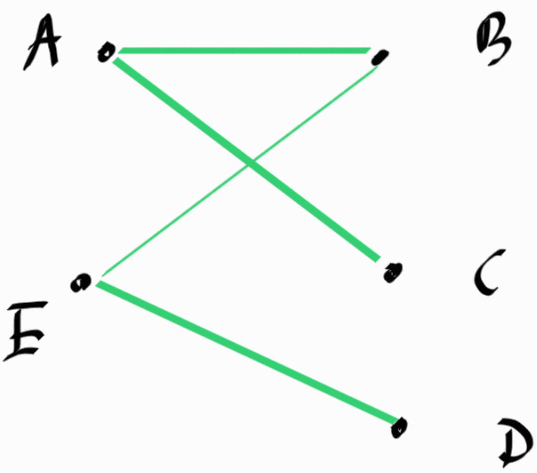
Def Bipartite graphs.

A graph G is called Bi-partite graph if $G = (L \cup R, E)$ s.t $L \cap R = \phi$ and $E \subseteq \{\{l,r\} : l \in L \wedge r \in R\}$



Basically, graph vertices should be able to partition into two sets such that these two sets are disjoint & all edges should go between nodes of L & nodes of R.

Ex:



$V = \{A, B, C, D, E\}$

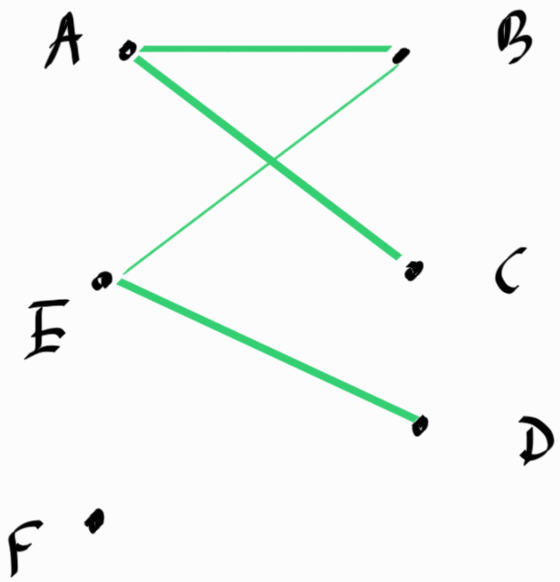
$L = \{A, E\}$

$R = \{B, C, D\}$

$L \cap R = \phi$

$E = \{\{A,B\}, \{A,C\}, \{E,B\}, \{E,D\}\}$

Ex:



$V = \{A, B, C, D, E, F\}$

$L = \{A, E, F\}$

$R = \{B, C, D\}$

$L \cap R = \emptyset$

$L' = \{A, E\}$

$R' = \{B, C, D, F\}$

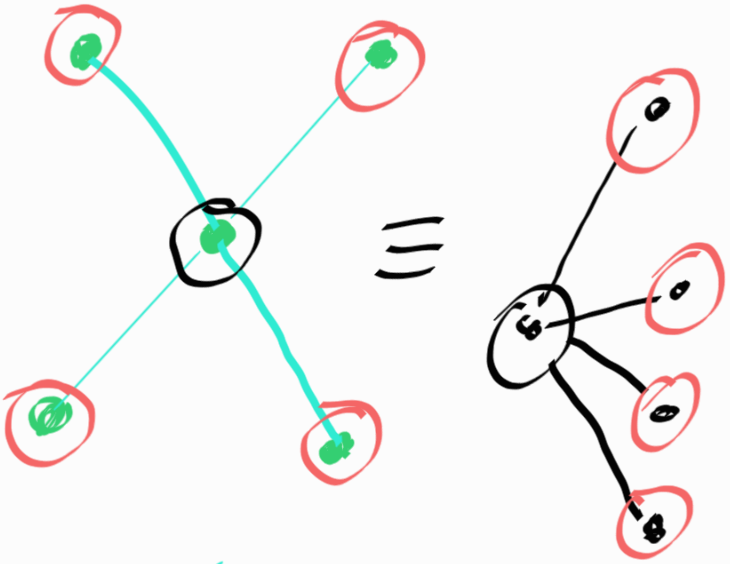
$L' \cap R' = \emptyset$

$L'' = \{A, B, E\}$

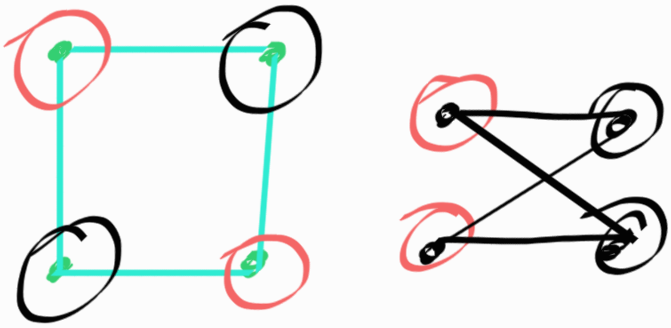
$R'' = \{C, D, F\}$

$L'' \cap R'' = \emptyset$

Popup test 09

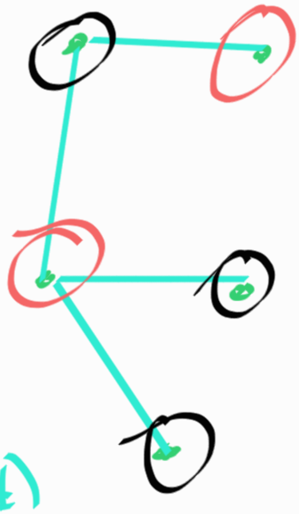


a) ✓



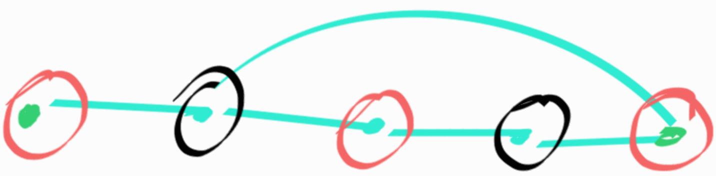
b) ✓

c) X



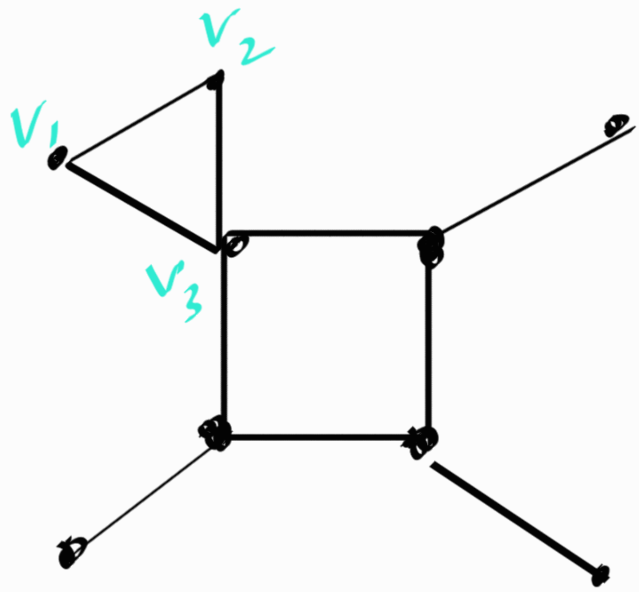
d) ✓

e)



✓

claim: Let G be an undirected graph. If G contains a triangle, then it is not a bipartite graph.



G contains a triangle $\Rightarrow$ G is <u>not</u> bipartite

$$p \Rightarrow q$$

$$\neg(p \Rightarrow q) \equiv p \wedge \neg q$$

Proof: Aiming for contradiction.

negation statement: G contains a triangle and G is bipartite.

Let v_1, v_2, v_3 be the nodes of a triangle in the graph G. without loss of generality, suppose $v_1 \in L$, $v_2 \in R$

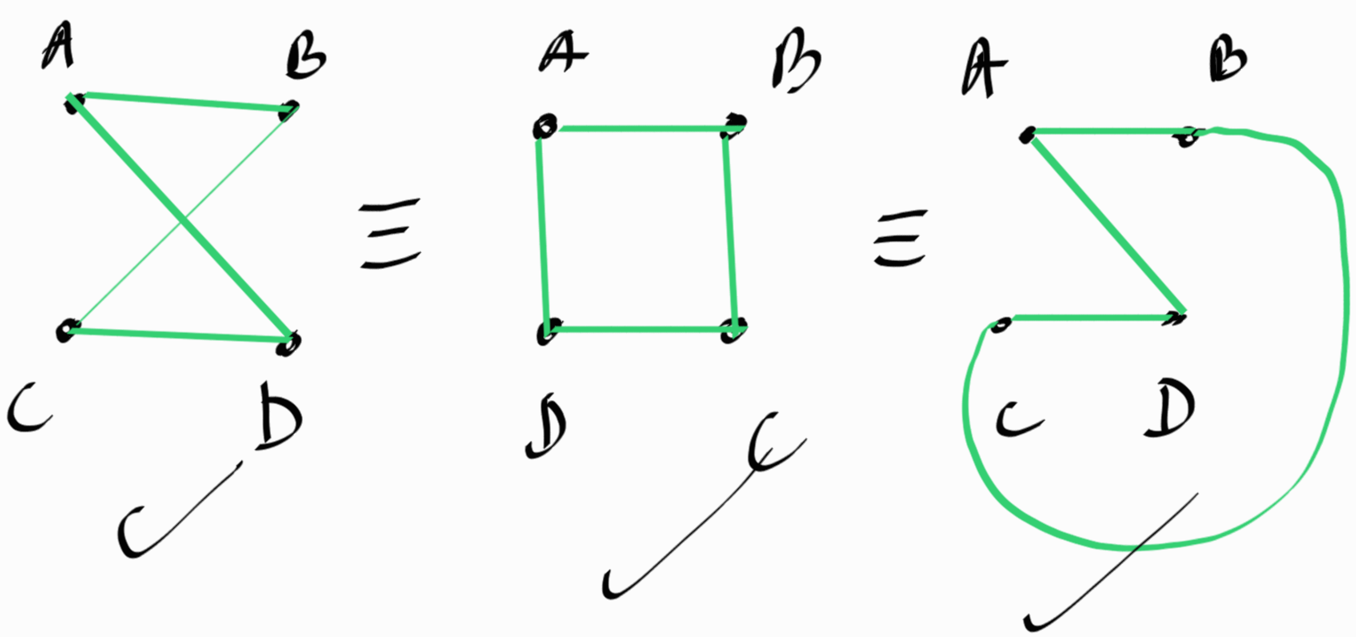
Since $V_2 \in R$, $V_3 \in L$, but there is an edge from V_1 to V_3 and both of them are in L.

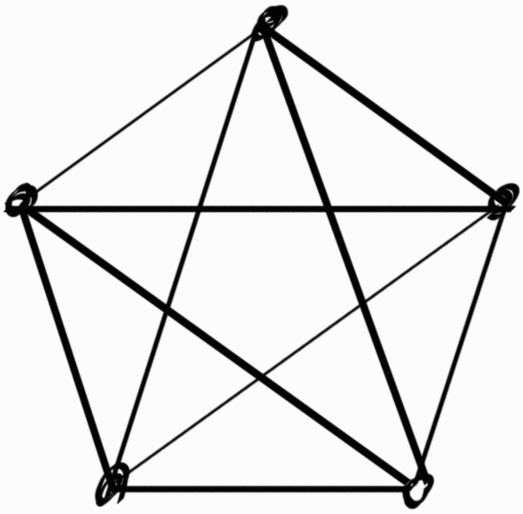
This is a contradiction.

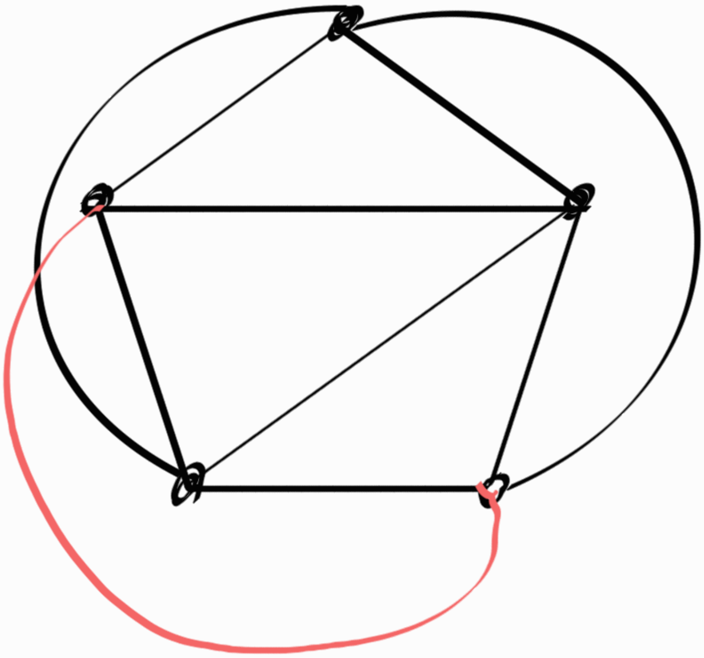
Therefore, initial claim is true $\blacksquare$

Def: A graph G is planar, if we can draw it in the 2D-plane without edge crossing.

Ex:



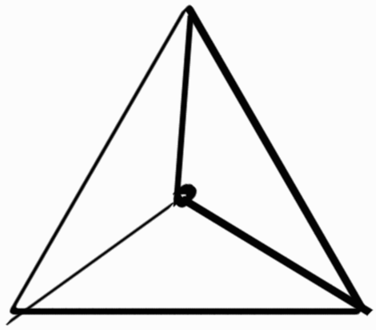




K_5

K_4

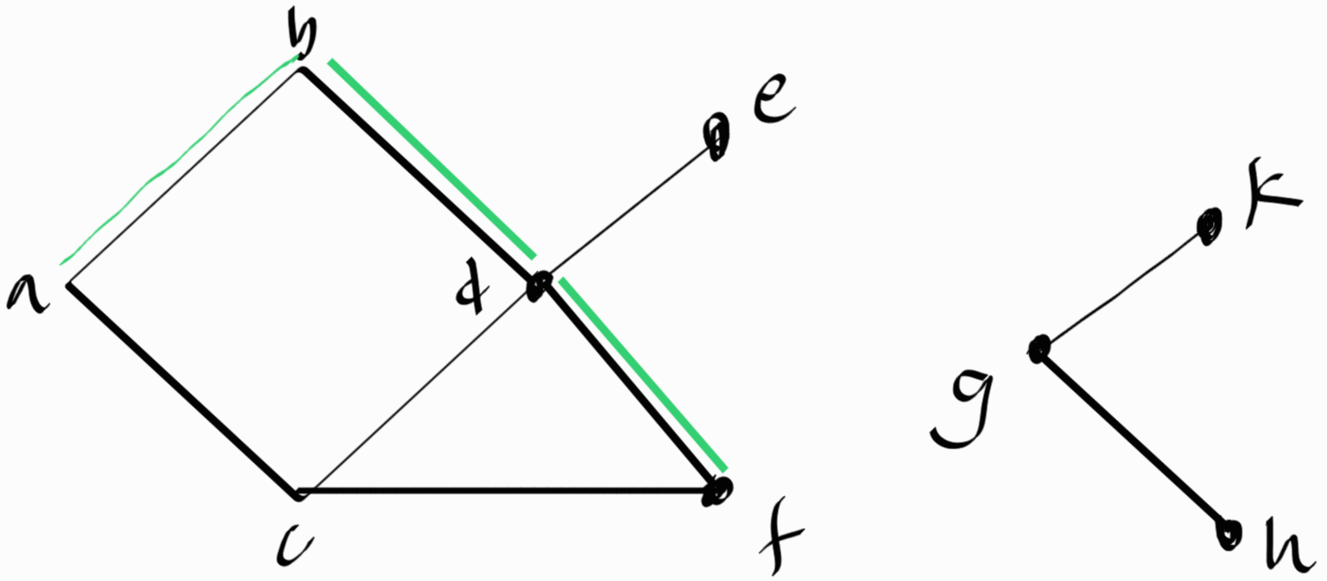
=



✓

Def: A path in a graph $G=(V,E)$ is a sequence of nodes $\langle u_1, u_2, u_3, \ldots, u_k \rangle$ s.t

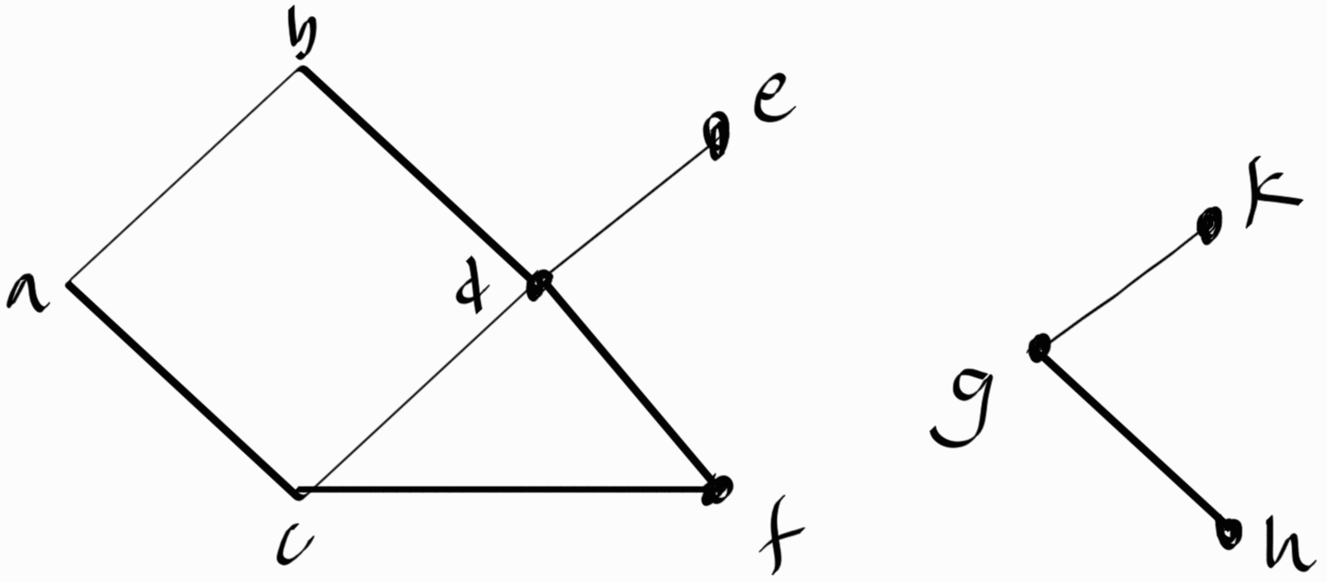
- $\forall i \in \{1,2,3,\ldots,k\}: u_i \in V$
- $\forall i \in \{1,2,3,\ldots,k-1\}: \{u_i, u_{i+1}\} \in E$



green path from $a \to f$: $\langle a,b,d,f \rangle$

is $\langle a,b,e \rangle$ a path in this graph? No

Def A path is simple if all of its nodes are unique.



⟨a, b, d, f, c, d, e⟩ is a valid path but not a simple path as the vertex d is repeated.

⟨a, b, d, f⟩ is a simple path

Def Length of a path

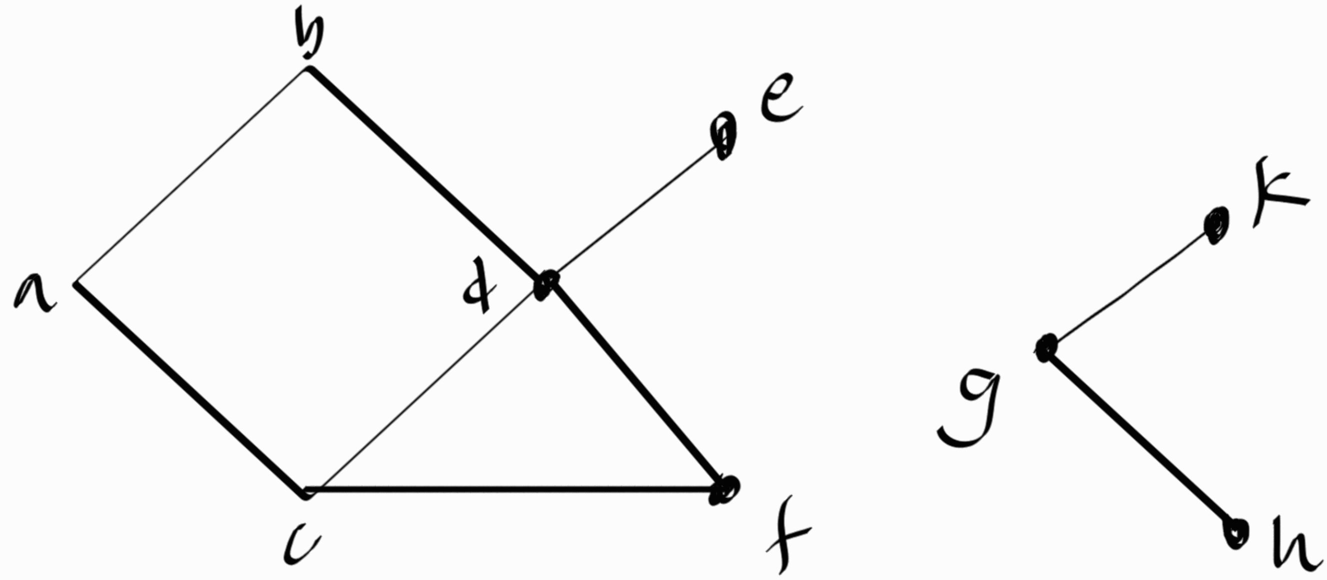
The length of a path is the number of edges in that path.

The length of $\langle a, b, d, f \rangle$ would be 3.

In general the length of the path $\langle u_1, u_2, u_3, \ldots\ldots, u_{k-1}, u_k \rangle$ is $k-1$.

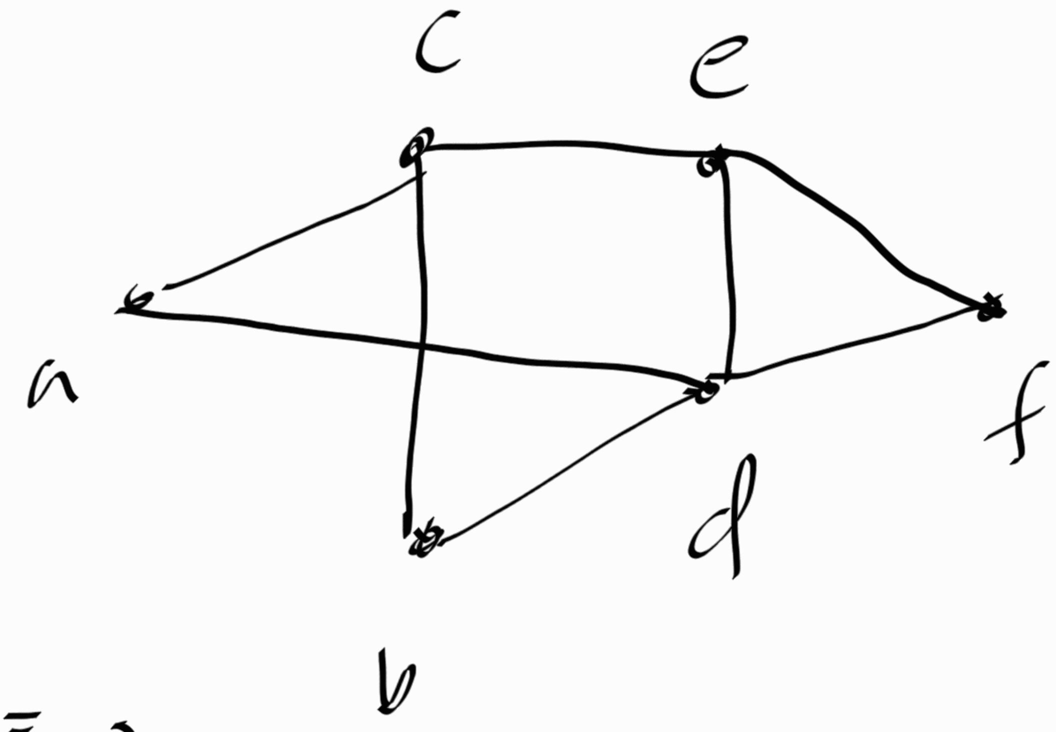
Def shortest path between two nodes.

Let $G=(V,E)$ be a graph (undirected or directed), and let $s \in V, t \in V$ be two nodes. A path from s to t is the shortest path if its length is the smallest out of all the paths from s to t.



The length of the shortest path from a to f is 2.

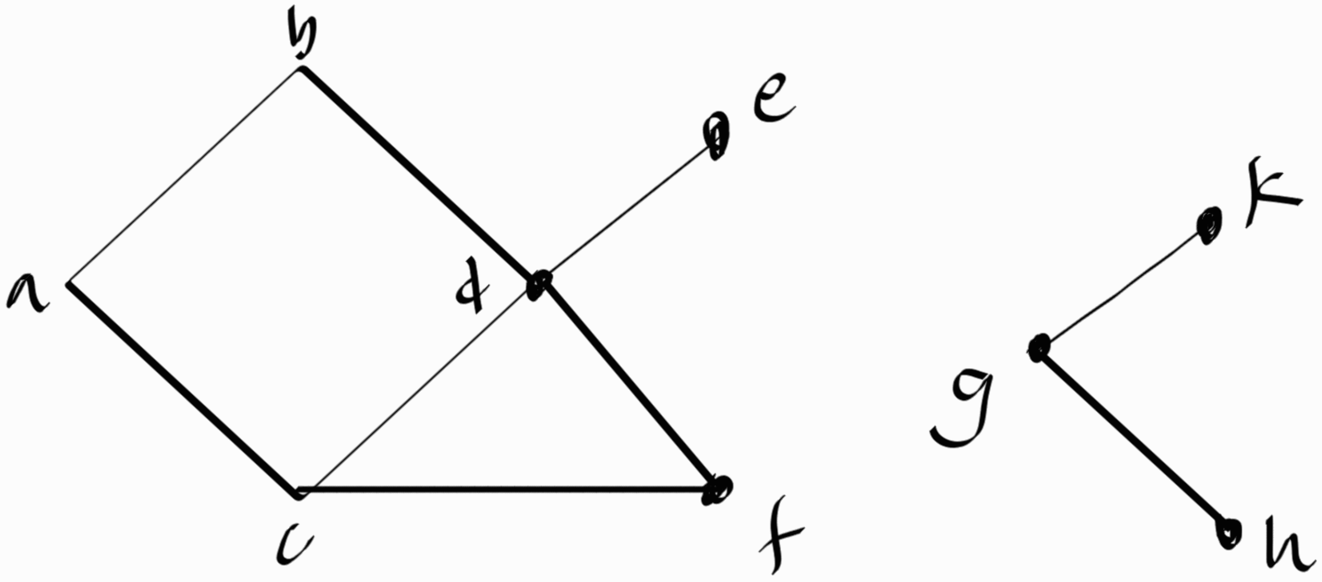
Def The distance, $dist(u,v)$, $d(u,v)$ between u and v is the length of the shortest path between u and v.



$d(a,f) = 2$

Def : A graph G is connected if

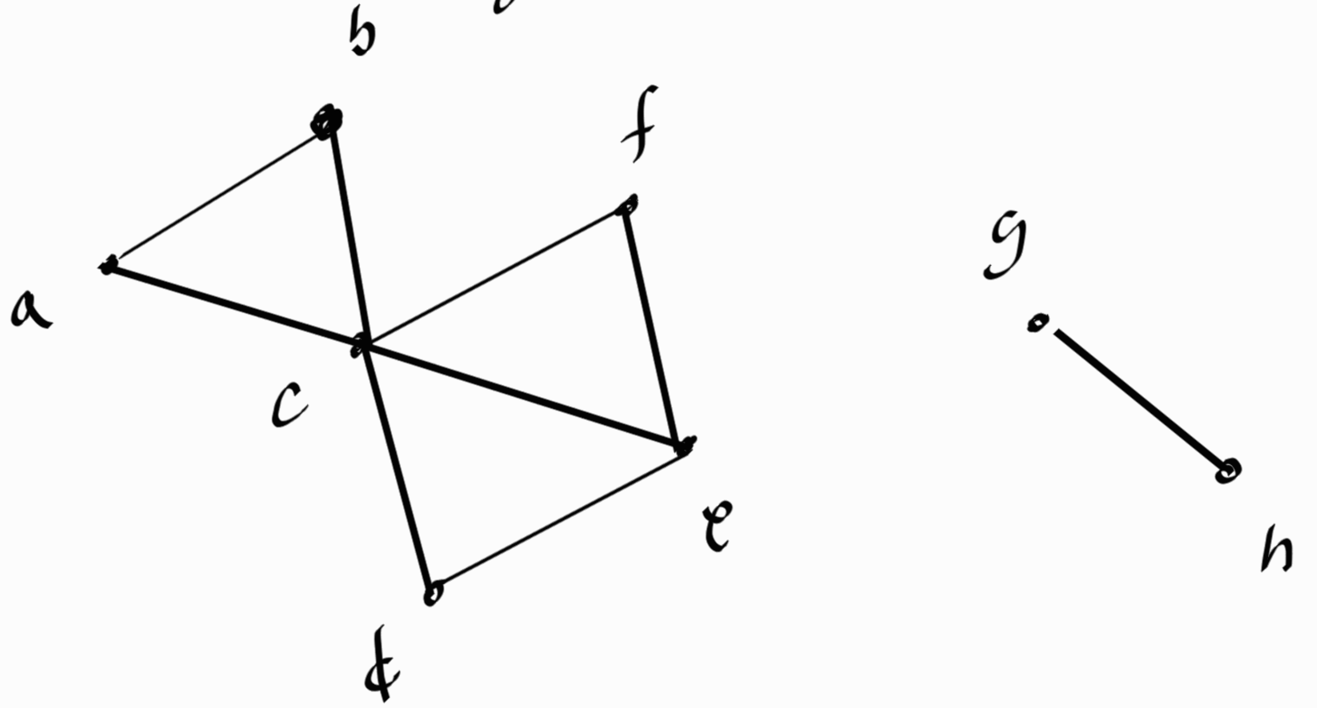
$\forall u, v \in V$, $\exists$ "a path" from u to v



G

This graph G is not connected because there is no path between e and k.

Def A cycle $\langle u_1, u_2, u_3, \ldots, u_k, u_1\rangle$ is a path of length ≥ 2 from u_1 to u_1 that does not traverse the same edge twice.

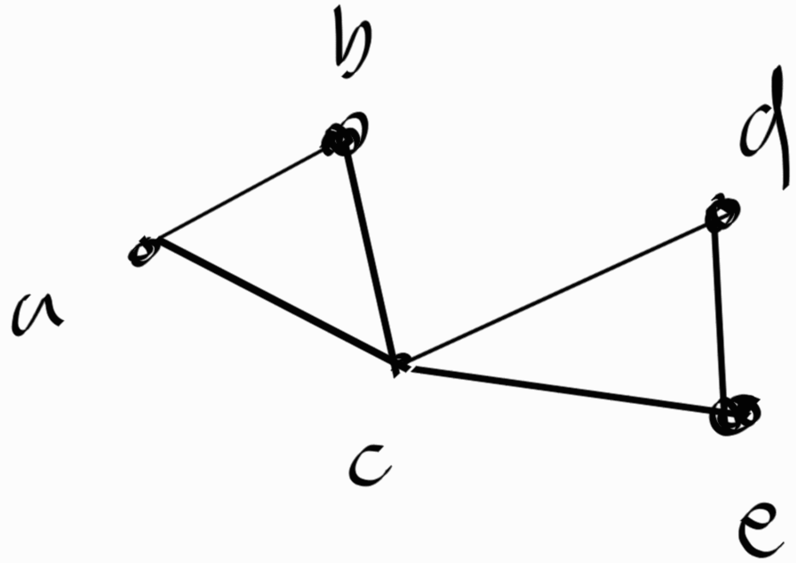


The path $\langle a, b, c, a\rangle$ ✓ (cycle)

The path $\langle a, b, c, f, e, c, a\rangle$ is a cycle ✓

The path $\langle a, b, c, f, e, c, b, a\rangle$ is not a cycle

Def A cycle is simple if its nodes are distinct.

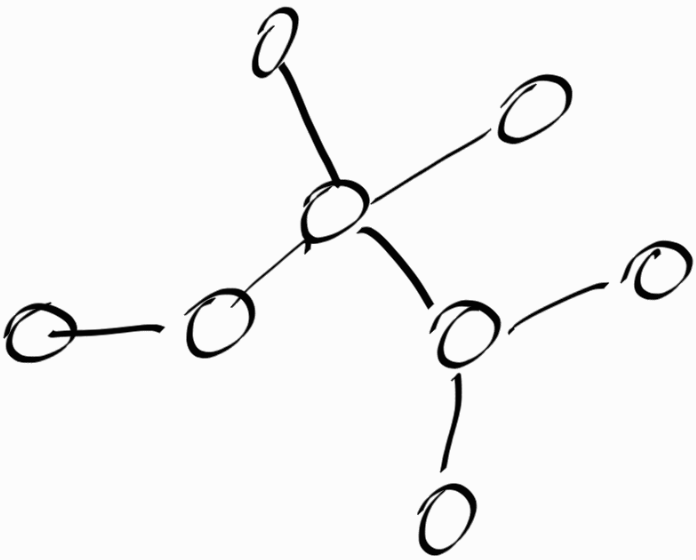


$\langle a, b, c, d, e, c, a \rangle$ is a cycle but not a simple cycle as c is traversed more than once.

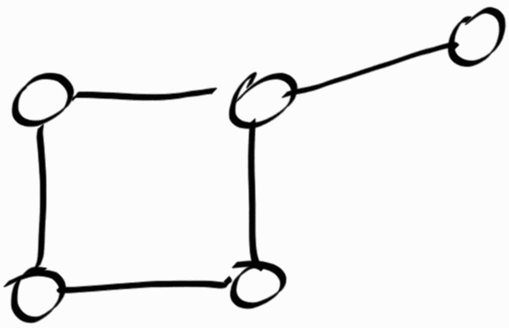
$\langle a, b, c, a \rangle$ is a simple cycle

Def A graph is acyclic if it contains no cycles.

Acyclic

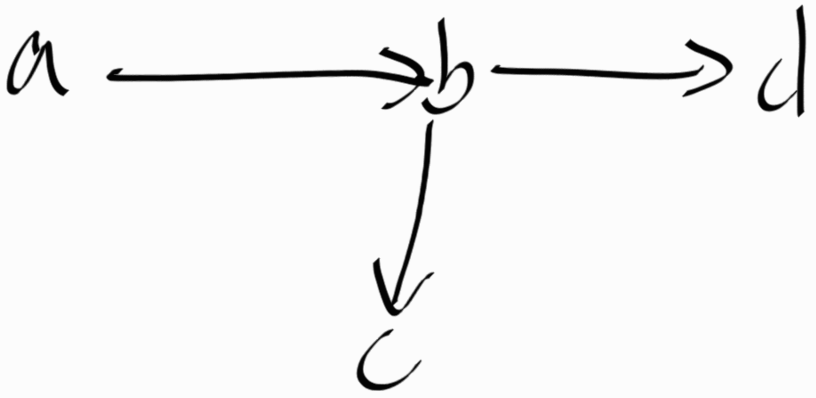


Acyclic

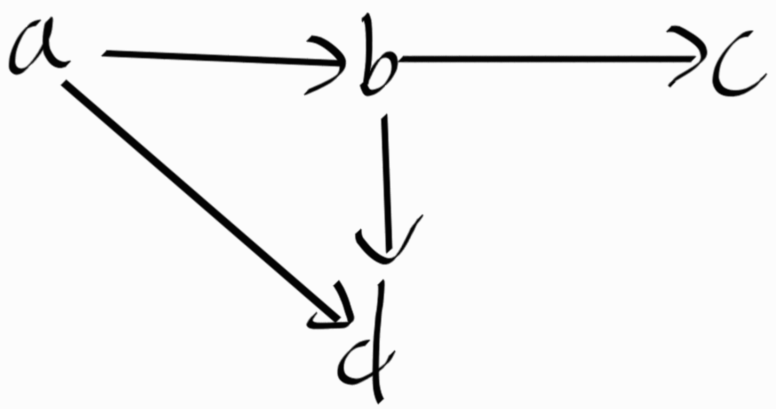


not acyclic

In directed graphs

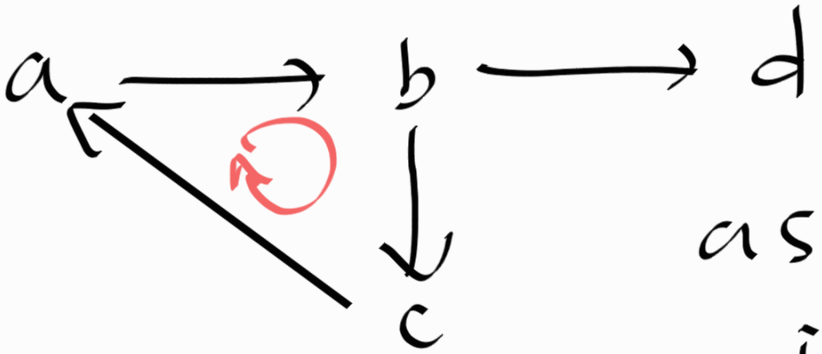


acyclic



acyclic

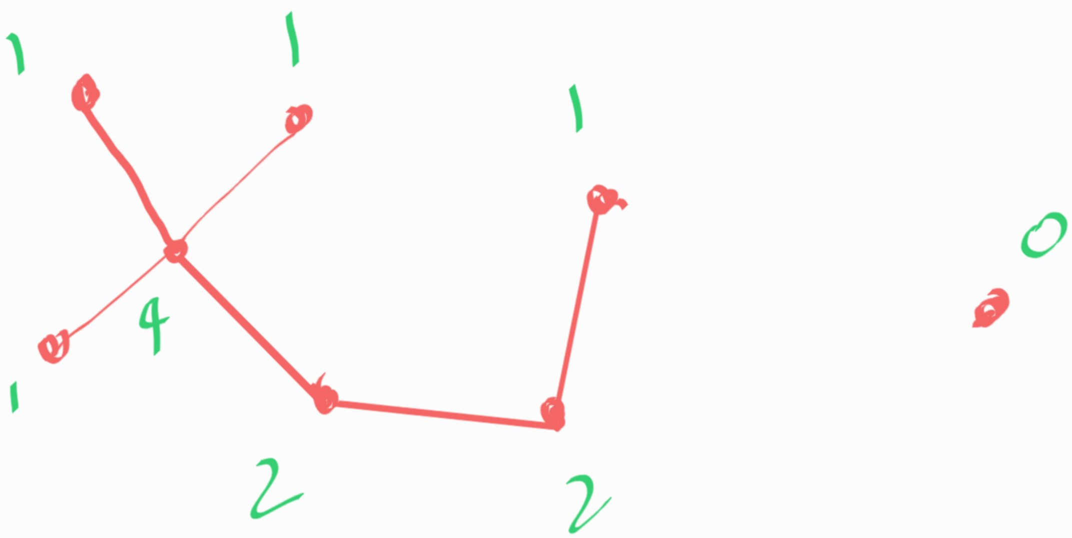
$\langle a, b, d, a \rangle$ is <ins>not</ins> a path in this directed graph.



not acyclic as $\langle a, b, c, a \rangle$ is a cycle.

lemma 11.3

If $G=(V,E)$ is an undirected acyclic graph, then $\exists v \in V$ s.t $\deg(v)=0$ or $\deg(v)=1$



Proof: we give a proof by construction.

Note: when we have a claim with $\exists$, one way of proving this claim is by showing we can always construct $\exists$ instance, We give a proof by constrution via an algorithm.

Algorithm:

Let u_0 be any node in V

Let $i = 0$

while current node u_i has unvisited neighbors:

Let u_{i+1} = any such unvisited neighbor

$i = i+1$

return u_i

Let t be the node returned by the algorithm on a graph G.

WTS: $\deg(t) = 0$ or $\deg(t) = 1$

case 1: let $t = u_0$, if $t = u_0$, that means there is no neighbors for node u_0, therefore, $\deg(u_0) = \deg(t) = 0$

Case 2: $t = u_k$, $k \geq 1$, we show that $\deg(t) = 1$

Since u_k is the last in the sequence $\langle u_1, u_2, u_3, \ldots, u_{k-1}, u_k \rangle$

there is no edge from $t = u_k$ to any other unvisited nodes.

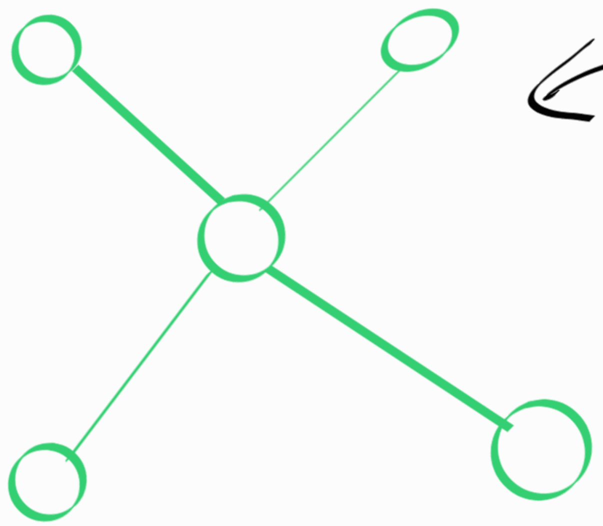
If $\exists$ some edge from t to any other node u_j other than u_{k-1}, then it has to be in $\langle u_0, u_1, u_2, \ldots\ldots, u_{k-2} \rangle$

But then it means $(u_j, \ldots\ldots u_{k-1}, u_k, u_j)$ is a cycle. But no such cycle exists in the graph.

Therefore $\deg(u_k) = \deg(t) = 1$

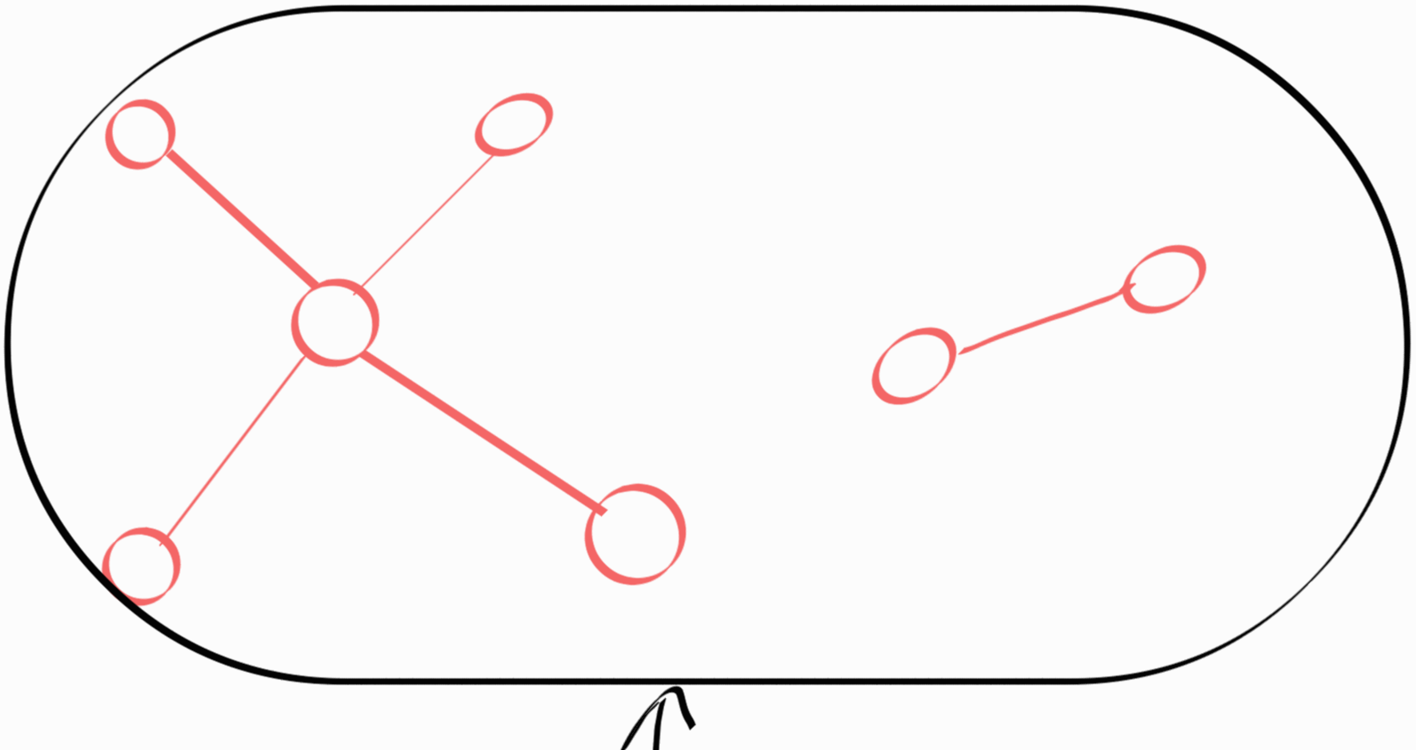
Def Tree

A tree is a graph that is connected and acyclic.

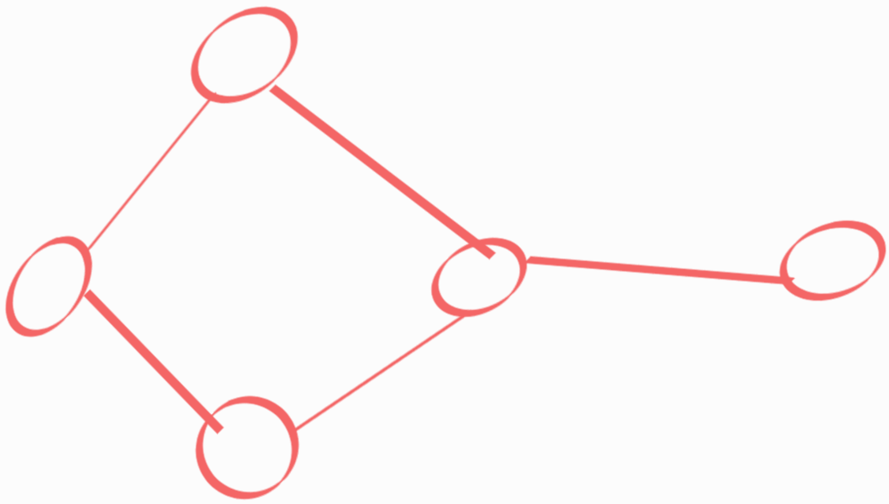


tree

not a tree



we call this a forrest.
(A set of trees)

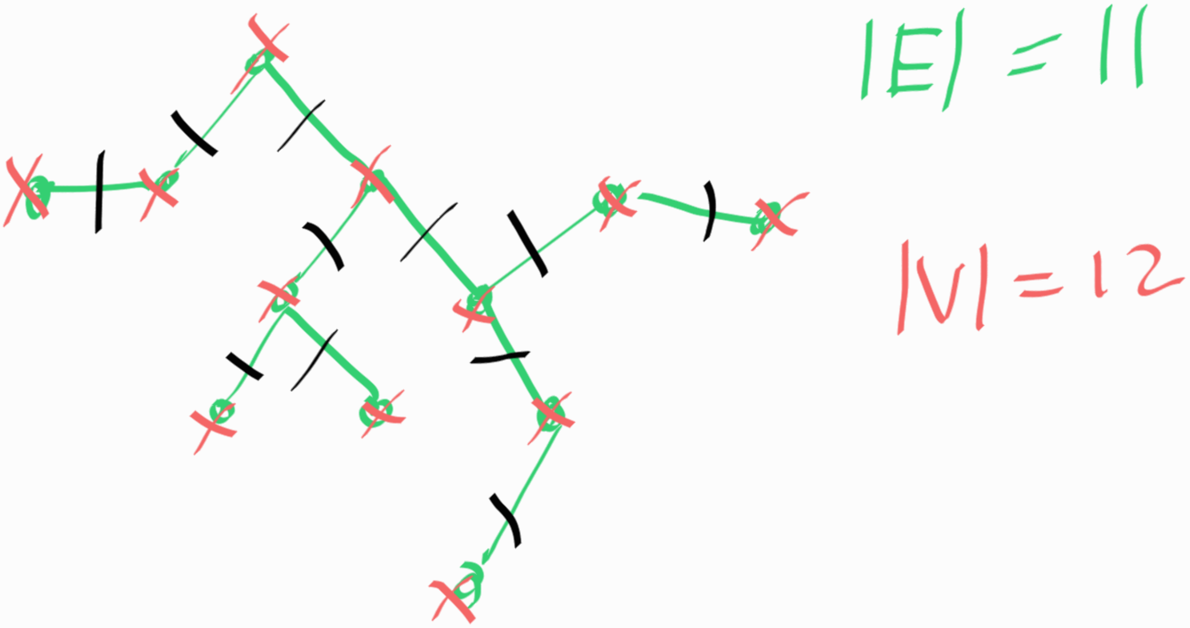


not a tree

Theorem

If $T=(V,E)$ is a tree, then

$$|E| = |V| - 1$$



Theorem

If $T = (V, E)$ is a tree,

then

1. Adding an edge creates a cycle
2. removing an edge disconnects graph.